INTRODUCTION

This manual for trade practical is intended for use in the ITI workshop. It consists of a series of practical exercises that are to be completed by the trainees during the first year of course is the **Electrician trade under Power Sector. It is National Skills Qualifications Framework NSQF Level - 4 (Revised 2022),** supplemented and supported by instructions/information to assist the trainees in performing the exercise. The exercises are designed to ensure that all the skills prescribed in the syllabus are covered including the allied trades. The syllabus for the 1st Year **Electrician** Trade under **Power Sector** Trade Practical is divided into Twelve Modules. The allocation of time for the various modules is given below:

Module 1 - Safety Practice and Hand Tools	40 Hrs
Module 2 - Wires, Joints - Soldering - U.G. Cables	95 Hrs
Module 3 - Basic Electrical Practice	51 Hrs
Module 4 - Magnetism and Capacitors	32 Hrs
Module 5 - AC Circuits	77 Hrs
Module 6 - Magnetism and Capacitors	50 Hrs
Module 7 - Basic Wiring Practice	110 Hrs
Module 8 - Wiring Installation and earthing	115 Hrs
Module 9 - Illumination	45 Hrs
Module 10 - Measuring Instruments	75 Hrs
Module 11 - Domestic Appliances	75 Hrs
Module 12 - Transformers	75 Hrs
Total Hrs	840 Hrs

The syllabus and the content in the modules are interlinked. As the number of workstations available in the electrical section is limited by the machinery and equipment, it is necessary to interpolate the exercises in the modules to form a proper teaching and learning sequence. The sequence of instruction is given in the schedule of instruction which is incorporated in the Instructor's Guide. With 25 practical hours a week of 5 working days 100 hours of practical per month is available.

Contents of Trade Practical

The procedure for working through the 106 exercises for the 1st Year with the specific objectives as the learning out comes at the end of each exercise is given is this book.

The skill objectives and tools/instruments, equipment/machines and materials required to perform the exercise are given in the beginning of each exercise. Skill training in the shop floor is planned through a series of practical exercises/experiments to support the related theory to make the trainees get hands on trainning in the Electrician trade along with the relevant cognitive skills appropriate for the level. A minimum number of projects have been included to make the training more effective and develop attitude to work in a team. Pictorial, schematic, wiring and circuit diagrams have been included in the exercises, wherever necessary, to assist the trainees broaden their views. The symbols used in the diagrams comply with the Bureau of Indian Standards (BIS) specifications.

Illustrations in this manual, help trainess visual perspective of the ideas and concepts. The procedures to be followed for completing the exercises is also given. Different forms of intermediate test questions have been included in the exercises, to enhance the trainee to trainee and trainee to instructor interactions.

Skill Information

Skill areas which are repetitive in nature are given as separate skill information sheets. Skills which are to be developed in specific areas are included in the exercises itself. Some subexercises are developed to fulfill the sequence of exercises in keeping with the syllabus.

This manual on trade practical forms part of the Written Instructional Material (WIM). Which includes manual on trade theory and assignment/test.

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